A condition for a local maximum of $n \mapsto w_{n,p}$

2026-04-17

Setting

Consider the all-heads-wins game with parameter $p \in (0,1)$: in each round, all remaining coins are tossed, at least one coin must be set aside, and the game is won iff every set-aside coin shows heads. Let $w_{n,p}$ denote the optimal winning probability starting with n coins, $w_{0,p} := 1$. By elementary conditioning on the number j of tails in the first round,

$$w_{n,p} \;=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \max_{j \le m \le n-1} w_{m,p}, \qquad n \ge 1. \tag{1}$$

Throughout, write

$$M_{n,j} \;:=\; \max_{j \le m \le n-1} w_{m,p} \qquad (1 \le j \le n-1).$$

Local maximum

Definition. We say that $n \mapsto w_{n,p}$ has a *strict local maximum* at $n^\star \ge 2$ if

$$w_{n^\star-1,p} \;<\; w_{n^\star,p} \;>\; w_{n^\star+1,p}.$$

Definitional inequalities

Directly from the definition, $n^\star$ is a strict local maximum iff

$$w_{n^\star,p} - w_{n^\star-1,p} > 0 \quad \text{and} \quad w_{n^\star,p} - w_{n^\star+1,p} > 0. \tag{2}$$

Reduction to strategy (B)

Let $b_{n,p}$ denote the winning probability under strategy (B) (set aside every head). It satisfies the same recursion (1) with the $\max_{j \le m \le n-1} w_{m,p}$ replaced by $b_{j,p}$:

$$b_{n,p} \;=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \, b_{j,p}, \qquad b_{0,p} := 1. \tag{3}$$

Lemma 1 (Reduction)**.** *If $(w_{m,p})_{m=1}^N$ is monotone non-increasing, then $w_{m,p} = b_{m,p}$ for every $m \in \{1, \dots, N\}$.*

Proof. Induction on m. For $m = 1$ both sides equal p. For $2 \le m \le N$, the monotonicity hypothesis gives $\max_{j \le m' \le m-1} w_{m',p} = w_{j,p} = b_{j,p}$ by the inductive hypothesis, so (1) at index m reduces exactly to (3) at index m. □ □

Corollary 2 (First non-monotonicity of w via b)**.** *Let $n^\star \ge 2$ be the smallest integer with $b_{n^\star,p} > b_{n^\star-1,p}$. Then $w_{m,p} = b_{m,p}$ for every $m \le n^\star$, and*

$$w_{n^\star,p} \;>\; w_{n^\star-1,p}. \tag{4}$$

In particular $n \mapsto w_{n,p}$ is not monotone non-increasing.

Proof. By the minimality of $n^\star$ in the statement (and by the equalities $w_{m,p} = b_{m,p}$ proved below for $m < n^\star$), $(b_{m,p})_{m=1}^{n^\star-1}$ is monotone non-increasing. If $(w_{m,p})_{m=1}^{n^\star-1}$ is also monotone non-increasing, Lemma 1 gives $w_{m,p} = b_{m,p}$ for $m < n^\star$; we verify this by induction. For $m = 1$ both equal p. Assume $w_{m',p} = b_{m',p}$ for $m' < m \le n^\star - 1$; then $b_{1,p} \ge \cdots \ge b_{m-1,p}$ translates to $w_{1,p} \ge \cdots \ge w_{m-1,p}$, so $M_{m,j} = w_{j,p} = b_{j,p}$ in (1), and (1) reduces to (3), giving $w_{m,p} = b_{m,p}$. The same argument at $m = n^\star$ gives $w_{n^\star,p} = b_{n^\star,p}$. In particular $w_{n^\star,p} = b_{n^\star,p} > b_{n^\star-1,p} = w_{n^\star-1,p}$. □ □

Remark. Corollary 2 does *not* claim that $n^\star$ is a strict local maximum of $n \mapsto w_{n,p}$. Indeed, at $p = 0.45$ the first b-uptick occurs at $n^\star = 7$ (with $b_7 \approx 0.4111$), while the first strict local maximum of w occurs only at $n = 15$; between $n^\star$ and 15 the sequence $w_{n,p}$ continues to increase (through further b-upticks). Pinning down the location of a local maximum requires running the Bellman recursion past $n^\star$.

Uniform upper bounds on $w_{n,p}$ for $p < \frac{1}{2}$

The Bellman value never exceeds $w_{1,p} = p$ for $n \geq 2$, and never exceeds $w_{2,p}$ for $n \geq 3$. Both bounds follow by direct induction on the Bellman.

Proposition 3 ($w_{n,p} < p$ for every $n \geq 2$, $p < \frac{1}{2}$)**.** *Let $p \in (0, \frac{1}{2})$ and $q := 1 - p$. Then for every $n \geq 2$,*

$$w_{n,p} \;\leq\; p - pq\big(q^{n-1} - p^{n-1}\big) \;<\; p \;=\; w_{1,p}, \tag{5}$$

with equality at $n = 2$.

Proof. By induction on n.

Base $n = 2$. Strategies (A) and (B) coincide on $n = 2$ and give $w_{2,p} = p^2(3 - 2p)$. Hence $p - w_{2,p} = p\big(1 - 3p + 2p^2\big) = p(1 - p)(1 - 2p) = pq(q - p) = pq(q^1 - p^1)$. This is positive for $p \in (0, \frac{1}{2})$ and matches (5) with equality.

Induction step ($n \geq 3$). Assume $w_{m,p} < p$ for every $m \in \{2, \ldots, n-1\}$, so that $\max_{j \leq m \leq n-1} w_{m,p} = w_{1,p} = p$ when $j = 1$ (the unique maximizer is $m = 1$) and $\max_{j \leq m \leq n-1} w_{m,p} < p$ when $j \geq 2$ (the range excludes 1). Substituting in the Bellman (1) and using strictness for $j \geq 2$ (the sum is non-empty since $n \geq 3$):

$$\begin{aligned} w_{n,p} \;&<\; p^n + \binom{n}{1} p^{n-1} q \cdot p + p \sum_{j=2}^{n-1} \binom{n}{j} p^{n-j} q^j \\ &=\; p^n + np^n q + p\big(1 - p^n - np^{n-1}q - q^n\big) \\ &=\; p + p^n q - pq^n \;=\; p - pq\big(q^{n-1} - p^{n-1}\big), \end{aligned}$$

where the second equality uses $\sum_{j=2}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - p^n - np^{n-1}q - q^n$ (from the binomial theorem). The right-hand side is $< p$ because $q > p$ for $p < \frac{1}{2}$, hence $q^{n-1} > p^{n-1}$. □ □

The same induction template, with $w_{2,p}$ in place of $w_{1,p}$, yields a strictly tighter bound.

Proposition 4 ($w_{n,p} < w_{2,p}$ for every $n \geq 3$, $p < \frac{1}{2}$)**.** *Let $p \in (0, \frac{1}{2})$. For every $n \geq 3$,*

$$w_{n,p} \;\leq\; w_{2,p} - p^2 q^2\, G_n(p) \;<\; w_{2,p}, \tag{6}$$

where

$$G_n(p) \;:=\; q^{\,n-2}(3 - 2p) - p^{\,n-2}\big[(n + 1) - 2p(n - 1)\big]. \tag{7}$$

Proof. Step 1: setup. By induction on n assume $w_{m,p} \leq w_{2,p}$ for $3 \leq m \leq n - 1$. Combined with Proposition 3 ($w_{m,p} < p = w_{1,p}$ for $m \geq 2$) the Bellman maxes split:

- $j = 1$: $\max_{1 \leq m \leq n-1} w_{m,p} = w_{1,p} = p$;
- $j = 2$: $\max_{2 \leq m \leq n-1} w_{m,p} = w_{2,p}$ (by the induction hypothesis, $w_{m,p} \leq w_{2,p}$ for $m \geq 3$);
- $j \geq 3$: $\max_{j \leq m \leq n-1} w_{m,p} \leq w_{2,p}$ (induction hypothesis, strict for $n \geq 4$).

Substituting in (1) and using $\sum_{j=2}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - p^n - np^{n-1}q - q^n$:

$$w_{n,p} \;\leq\; \bar{w}_n \;:=\; p^n(1 + nq) + w_{2,p}\big(1 - p^n - np^{n-1}q - q^n\big).$$

Step 2: reduction to $G_n(p) > 0$. Using $p - w_{2,p} = pq(1 - 2p)$ and grouping, $\bar{w}_n - w_{2,p} = -p^2 q^2\, G_n(p)$, so $\bar{w}_n < w_{2,p}$ iff $G_n(p) > 0$.

Step 3: $G_n(p) > 0$ on $(0, \frac{1}{2})$ for $n \geq 3$. By a second induction on n, with the algebraic identity

$$G_n(p) - q\, G_{n-1}(p) \;=\; p^{\,n-3}(1 - 2p)\big[3p + n(q - p)\big] \qquad (n \geq 4), \tag{8}$$

which is verified directly. For $p \in (0, \frac{1}{2})$, both factors $(1 - 2p)$ and $[3p + n(q - p)]$ are positive. Base case: $G_3(p) = q\big[(3 - 2p) - 4p\big] = 3q(1 - 2p) > 0$. Inductive step: $G_n(p) = qG_{n-1}(p) + p^{n-3}(1 - 2p)[3p + n(q - p)] > qG_{n-1}(p) > 0$.

Step 4: conclude. $w_{n,p} \leq \bar{w}_n = w_{2,p} - p^2 q^2 G_n(p) < w_{2,p}$. □ □

Remark. The cancellation in Step 2 — which collapses the $\binom{n}{2}p^{n-2}q^2$ terms — is what allows the induction to bootstrap from $w_{2,p}$ rather than just from $w_{1,p} = p$. The recursion (8) is the algebraic engine: it says $G_n - qG_{n-1}$ equals an explicitly positive quantity on $(0, \frac{1}{2})$, so the strict positivity of G_n propagates upward in n.

Empirically, for every tested $p \in (0.05, 0.5)$ at finite horizon $n_{\max} \leq 300$, the Bellman sequence has many strict local maxima, all with values $w_{N,p} < w_{2,p}$ — consistent with Propositions 3–4. The set of running-maximum indices of $(w_{n,p})_{n\geq 1}$ is exactly $\{1\}$ for every tested $p \in (0, \frac{1}{2})$.

Attempt to extend to $w_{n,p} < w_{3,p}$

The natural next step is the bound $w_{n,p} < w_{3,p}$ for $n \geq 4$, $p < \frac{1}{2}$. Numerically the bound holds for every tested (n, p). The same induction template — but now with a fourth bucket $j = 3$ — gives, after grouping,

$$\bar{w}_n - w_{3,p} \;=\; p^3 q^2\,(2p-1)\,R_n(p), \tag{9}$$

where R_n is an explicit polynomial in p of degree n with $R_n(0) = 13$ and $R_n(\frac{1}{2}) \geq 0$. The bound $\bar{w}_n \leq w_{3,p}$ is therefore equivalent (since $(2p-1) < 0$ on $(0, \frac{1}{2})$) to $R_n(p) > 0$ on $(0, \frac{1}{2})$.

Proposition 5 (partial result on $w_n < w_3$)**.** *For every $p \in (0, \frac{1}{2})$ and every $n \in \{4, \dots, 12\}$, $w_{n,p} < w_{3,p}$.*

Sketch. Same as Proposition 4: the Bellman maxes split as $j = 1 \Rightarrow w_1$, $j = 2 \Rightarrow w_2$, $j = 3 \Rightarrow w_3$, $j \geq 4 \Rightarrow$ value bounded by w_3 via the induction hypothesis. Substituting and identifying the sum $\sum_{j=3}^{n-1}\binom{n}{j}p^{n-j}q^j = 1 - p^n - q^n - np^{n-1}q - \binom{n}{2}p^{n-2}q^2$ yields (9). The polynomial R_n is then verified to be positive on the entire interval $(0, \frac{1}{2})$ by direct real-roots computation for each $n \in \{4, \dots, 12\}$. □

Remark (Breakdown for $n \geq 13$ near $p = \frac{1}{2}$). Unlike G_n in Proposition 4, the polynomial R_n does *not* satisfy a clean positivity-preserving recursion of the form $R_n - qR_{n-1} =$ (positive). Numerically, for $n \geq 13$ the polynomial R_n acquires a single real root $p_n \in (0, \frac{1}{2})$ that drifts towards $\frac{1}{2}$ from below as n grows ($p_{13} \approx 0.4978$, $p_{20} \approx 0.4911$, $p_{24} \approx 0.4900$). Hence the bound $\bar{w}_n \leq w_{3,p}$ *fails* on the subinterval $(p_n, \frac{1}{2})$ for $n \geq 13$, even though numerically the sharper claim $w_{n,p} < w_{3,p}$ continues to hold throughout. Closing the gap for $n \geq 13$ requires a tighter induction (e.g. chaining further bounds $w_{m,p} \leq w_{4,p}, w_{5,p}, \dots$) to refine the $j \geq 4$ contributions.

A tool for finding local maxima

Corollary 2 reduces the problem of locating a local maximum of the *Bellman* value function w to the (much simpler) problem of detecting non-monotonicity of the *strategy-(B)* sequence b. Strategy (B) has no max in its recursion (3); it is linear in the previous values. We record a few identities and structural observations that make the tool tangible.

Throughout, set $q := 1 - p$.

A probabilistic identity

Proposition 6 (Binomial-mixture identity)**.** *For every $n \geq 1$ and $p \in (0, 1)$,*

$$(1 + q^n)\,b_{n,p} \;=\; \mathbb{E}\big[b_{J,p}\big], \qquad J \sim \text{Binomial}(n, q). \tag{10}$$

Equivalently, $\mathbb{E}\big[b_{J,p} - b_{n,p}\big] = q^n b_{n,p}$.

Proof. Starting from the recursion (3) with $b_{0,p} = 1$ and noting that the $j = n$ term in the binomial expansion equals $q^n b_{n,p}$,

$$\begin{aligned} b_{n,p} &= \sum_{j=0}^{n-1}\binom{n}{j}p^{n-j}q^j\,b_{j,p} \;=\; \sum_{j=0}^{n}\binom{n}{j}p^{n-j}q^j\,b_{j,p} - q^n b_{n,p} \\ &= \mathbb{E}\big[b_{J,p}\big] - q^n b_{n,p}. \quad \square \end{aligned}$$

Polynomial structure

Writing $b_{n,p} = p^n P_n(q)$, the recursion (3) becomes

$$P_n(q) \;=\; \sum_{j=0}^{n-1}\binom{n}{j}q^j\,P_j(q), \qquad P_0 := 1. \tag{11}$$

Proposition 7. *$P_n(q)$ is a polynomial in q of degree $\binom{n}{2}$ with* non-negative integer *coefficients. Moreover*

$$P_n(0) = 1, \qquad P_n(1) = F_n, \qquad P_n(\tfrac{1}{2}) = 2^{\,n-1},$$

where $F_n = 1, 1, 3, 13, 75, 541, \ldots$ is the n-th Fubini number (ordered Bell number, OEIS A000670*).*

Proof. The recursion (11) is linear with non-negative integer coefficients and $P_0 = 1$, so induction gives non-negative integer coefficients. The degree recursion $\deg P_n = (n-1) + \deg P_{n-1}$, with $\deg P_1 = 0$, gives $\deg P_n = \binom{n}{2}$.

$P_n(0)$: only the $j = 0$ term contributes at $q = 0$, giving $\binom{n}{0} P_0(0) = 1$.

$P_n(1)$: at $q = 1$, (11) becomes $P_n(1) = \sum_{j=0}^{n-1} \binom{n}{j} P_j(1)$, which is the defining recursion of the Fubini numbers (with $F_0 = 1$).

$P_n(1/2)$: $b_{n,1/2} = \frac{1}{2}$ for all $n \geq 1$, so $P_n(1/2) = b_{n,1/2}/(1/2)^n = 2^{\,n-1}$. □ □

An exponential generating function

Proposition 8. *Let $G(x) := \sum_{n\geq 0} b_{n,p}\, x^n/n!$. Then*

$$G(x) \;=\; 1 + (e^{px} - 1)\, G(qx) \;=\; \sum_{k\geq 0} \prod_{i=0}^{k-1} \big(e^{pq^i x} - 1\big). \tag{12}$$

Proof. Multiply the recursion (3) by $x^n/n!$ and sum over $n \geq 1$; exchange orders of summation and identify $\sum_{n>j} (px)^{n-j}/(n-j)! = e^{px} - 1$. Iterating $G(x) = 1 + (e^{px} - 1)G(qx)$ and noting that the remainder term $\prod_{i=0}^{k-1} (e^{pq^i x} - 1)\, G(q^k x) \to 0$ as $k \to \infty$ (since $e^{pq^i x} - 1 \sim pq^i x$ for large i), the product expansion follows. □ □

Small-p heuristic

At leading order in $p \to 0^+$, (11) evaluated at $q \to 1$ gives $P_n(1) = F_n$ and hence

$$b_{n,p} \;\sim\; F_n\, p^n \qquad (p \to 0^+,\ n \text{ fixed}).$$

Fubini numbers satisfy $F_n/F_{n-1} \to n/\log 2$, so

$$\frac{b_{n,p}}{b_{n-1,p}} \;\sim\; p \cdot \frac{F_n}{F_{n-1}} \;\sim\; \frac{np}{\log 2} \qquad (p \to 0^+,\ n \text{ fixed}).$$

This formally predicts $b_{n,p} > b_{n-1,p}$ for $n > (\log 2)/p$. The prediction is *qualitatively* right but quantitatively much too optimistic: the Taylor expansion in p is pointwise (fixed n, $p \to 0$), whereas the first monotonicity failure $n_{\text{fail}}(p)$ diverges with n and the corrections of order $p^{n+1}, p^{n+2}, \ldots$ cannot be ignored. Empirically $n_{\text{fail}}(p) \asymp p^{-2.2}$, not p^{-1}.

Toward a general proof

Reformulation

Combining the tool of Corollary 2 with Proposition 7, our open conjecture admits the following purely analytic reformulation.

Conjecture 9. *For every $p \in (0, \frac{1}{2})$, the sequence $(b_{n,p})_{n\geq 1}$ is* not *monotone non-increasing in n. Equivalently, for every $q \in (\frac{1}{2}, 1)$ there exists $n \geq 2$ such that*

$$(1 - q)\, P_n(q) \;>\; P_{n-1}(q), \tag{13}$$

where P_n is the polynomial of Proposition 7.

By Corollary 2, Conjecture 9 implies that $n \mapsto w_{n,p}$ has a strict local maximum for every $p \in (0, \frac{1}{2})$.

What is known

We collect what can be proved toward Conjecture 9.

Boundary. At $q = \frac{1}{2}$ (i.e. $p = \frac{1}{2}$), Proposition 7 gives $(1-q)P_n(q) = \frac{1}{2} \cdot 2^{n-1} = 2^{n-2} = P_{n-1}(\frac{1}{2})$ for every $n \geq 2$, so (13) is an equality — the symmetric case is the boundary at which (B) is marginally monotone ($b_{n,1/2} = \frac{1}{2}$ for all n).

At $q = 1$. $P_n(1) = F_n$, Fubini numbers, and $F_n/F_{n-1} \to \infty$, so $(1-q)P_n(q)/P_{n-1}(q) \to \infty$ as $q \uparrow 1$ for every $n \geq 2$; the inequality (13) is (trivially) satisfied in a suitable limit.

Known specific p. For $p = 21/50$, the exact rational calculation of [1] gives $w_9 > w_8$; by Corollary 2 this is equivalent to $b_{9,21/50} > b_{8,21/50}$, i.e. (13) holds at $q = 29/50$, $n = 9$.

Asymptotic limit of b. If $(b_{n,p})$ were monotone non-increasing for all $n \geq 1$, then $b_{n,p} \downarrow L$ for some $L \geq 0$, and Proposition 6 together with bounded convergence gives $(1+q^n)b_{n,p} = \mathbb{E}[b_{J,p}] \to L$; the limit relation is therefore self-consistent (no direct contradiction).

A reduced identity for the forward difference

Let $\Delta b_n := b_{n,p} - b_{n-1,p}$. From (10) at indices n and $n-1$,

$$\begin{aligned}(1+q^n)b_{n,p} - (1+q^{n-1})b_{n-1,p} &= \mathbb{E}\big[b_{J_n,p}\big] - \mathbb{E}\big[b_{J_{n-1},p}\big] \\ &= q\,\mathbb{E}\big[b_{J_{n-1}+1,p} - b_{J_{n-1},p}\big] \\ &= q\,\mathbb{E}\big[\Delta b_{J_{n-1}+1}\big],\end{aligned}$$

using the coupling $J_n \overset{d}{=} J_{n-1} + \mathrm{Bern}(q)$. The left-hand side expands as $\Delta b_n + q^{n-1}(qb_{n,p} - b_{n-1,p})$, so

$$\boxed{\Delta b_n \;=\; q\,\mathbb{E}\big[\Delta b_{J_{n-1}+1}\big] + q^{n-1}\big(b_{n-1,p} - q\,b_{n,p}\big).} \tag{14}$$

Identity (14) expresses the target quantity Δb_n as a *binomial-weighted average* of earlier $\Delta b_\bullet$'s plus an explicit positive correction of order q^{n-1}. If $(b_{n,p})$ were monotone non-increasing, then $\Delta b_\bullet \leq 0$, the first term of (14) would be ≤ 0, and the second would be $q^{n-1}(b_{n-1,p} - qb_{n,p}) \geq q^{n-1}(1-q)b_{n-1,p} > 0$.

Thus Conjecture 9 reduces to the following.

Conjecture 10 (equivalent to Conjecture 9)**.** *For every $p \in (0, \frac{1}{2})$ there exists n such that*

$$q\,\mathbb{E}\big[(-\Delta b_{J_{n-1}+1})\big] \;<\; q^{n-1}\big(b_{n-1,p} - q\,b_{n,p}\big).$$

Both sides of the inequality tend to 0, but the right-hand side decays like $q^{n-1} \cdot p \cdot b_{n-1,p}$ (still assuming monotone non-increasing, so $b_{n-1,p} \geq L \geq 0$), whereas the left-hand side is a binomial average of past decrements; a precise rate comparison is the missing step.

Status

Conjecture 9 — and hence the analytic existence of a strict local maximum of $n \mapsto w_{n,p}$ for every $p \in (0, \frac{1}{2})$ — remains open. The attack via (14) is the cleanest route so far: it requires only control on the *sizes* of past decrements $|\Delta b_j|$ weighted by a binomial measure, compared against an explicit exponentially small positive term. A quantitative estimate of the form

$$\sum_{j=1}^{n-1} \binom{n-1}{j} p^{\,n-1-j} q^j \cdot |\Delta b_j| \;=\; o\big(q^{n-1} \cdot p \cdot b_{n-1,p}\big)$$

along some subsequence $n \to \infty$ would close the argument.

References

[1] *An explicit (n,k) at which the optimal strategy is neither (A) nor (B)*, `Manuscript/proof_intermediate_p042.tex`, 2026.